Insight-V: Exploring Long-Chain Visual Reasoning with Multimodal Large Language Models

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Abstract

Multimodal Large Language Models (MLLMs) have shown remarkable progress in visual understanding, but they still struggle with complex, long-chain visual reasoning tasks that require step-by-step analysis and inference. This limitation stems from the scarcity of high-quality long-chain reasoning data and ineffective training paradigms that entangle reasoning with answering, leading to error propagation and instability. To address these challenges, we introduce ***Insight-V****, a novel framework that couples scalable generation of structured visual reasoning traces with a multi-agent training paradigm. Our approach separates reasoning from summarization through specialized agents: a* reasoning agent *that generates detailed step-by-step analysis, and a* summary agent *that produces final answers based on the reasoning chain. We generate approximately 200K high-quality long-chain reasoning paths using a progressive data generation pipeline with multi-granularity assessment, then refine the models through iterative Direct Preference Optimization (DPO). Extensive experiments on 7 visual reasoning benchmarks demonstrate significant improvements: Insight-V enhances LLaVA-NeXT-8B by +7.0 points average (40.2→47.2), with notable gains on MMMU (36.9→42.0), MMMU-Pro (13.2→21.0), and MMBench (72.3→81.7). Our approach also maintains strong general perception capabilities, showing improvements on TextVQA, DocVQA, and OCRBench. The results demonstrate that decomposing visual reasoning and training on high-quality long-chain data substantially advances MLLMs' reasoning capabilities without sacrificing perception.*

1. Introduction

Multimodal Large Language Models (MLLMs) have achieved remarkable success in understanding and reasoning about visual content [**?** **?**]. However, despite their impressive performance on many vision-language tasks, they still face significant challenges when tackling complex visual reasoning problems that require long chains of step-by-step analysis. Tasks such as mathematical problem solving with diagrams, scientific reasoning with charts and graphs, and multi-step logical inference from visual scenes demand not only accurate perception but also systematic reasoning capabilities that current MLLMs often lack.

The core challenges in developing MLLMs with strong long-chain visual reasoning capabilities are threefold. First, there is a severe scarcity of high-quality training data that demonstrates detailed, step-by-step visual reasoning processes. Most existing multimodal datasets focus on direct question-answering without providing intermediate reasoning steps, making it difficult for models to learn systematic reasoning patterns [**?** **?**]. Second, conventional training paradigms directly supervise models to generate both reasoning and final answers simultaneously, which often leads to instability and error propagation where mistakes in early reasoning steps cascade through the entire chain. Third, the entanglement of reasoning and answering processes makes it challenging to isolate and improve specific reasoning capabilities without affecting overall performance.

To address these fundamental challenges, we propose **Insight-V**, a comprehensive framework for enhancing long-chain visual reasoning in MLLMs. Our key insight is to decompose the visual reasoning process into two specialized components: a *reasoning agent* that focuses on generating detailed, structured reasoning chains, and a *summary agent* that produces final answers based on the reasoning content. This separation allows each agent to be optimized for its specific role while providing robustness against error propagation.

Insight-V consists of three main components: (1) a scalable data generation pipeline that creates high-quality long-chain visual reasoning traces using progressive generation and multi-granularity assessment, (2) a multi-agent training framework that separately optimizes reasoning and summarization capabilities, and (3) an iterative preference optimization procedure that refines the quality of reasoning chains through Direct Preference Optimization (DPO) [**?**].

Our data generation pipeline employs a progressive reasoning generator that produces structured JSON-formatted reasoning steps with explicit action indicators (continue/summary), enabling diverse reasoning path exploration for each query. We then apply multi-granularity assessment using both strong language models for answer verification

and multimodal models for reasoning quality scoring, resulting in approximately 200K high-quality reasoning chains.

The multi-agent training framework trains the reasoning agent on the best reasoning paths through supervised fine-tuning and iterative DPO refinement, while the summary agent learns to utilize or appropriately ignore the reasoning content passed through special <thoughts> tags. This design provides robustness to flawed reasoning chains while maintaining strong perception capabilities through mixed training on optimal paths, suboptimal paths, and general question-answering data.

We evaluate Insight-V extensively on seven visual reasoning benchmarks, including MMMU [?], MMMU-Pro [?], MMBench, ChartQA [?], MMStar, MathVista [?], and MME. Applied to LLaVA-NeXT-8B [?], our approach achieves an average improvement of +7.0 points (40.2→47.2), with substantial gains across all benchmarks. On our stronger 7B baseline model, we observe consistent improvements of +2.9 points average (48.7→51.6). Importantly, general perception capabilities remain strong or even improve, as demonstrated on TextVQA, DocVQA, OCRBench, and AI2D.

Our main contributions are: (1) A scalable, automated pipeline for generating high-quality long-chain visual reasoning data with progressive generation and multi-granularity assessment. (2) A multi-agent MLLM framework that disentangles reasoning and summarization, improving robustness to flawed reasoning chains. (3) An iterative DPO procedure that stabilizes and enhances long-chain reasoning quality using model-generated preferences. (4) Comprehensive experimental validation showing consistent improvements across diverse visual reasoning and perception benchmarks.

2. Related Work

Multimodal Large Language Models. Recent years have witnessed rapid progress in MLLMs, with architectures typically combining vision encoders with large language models [? ? ?]. LLaVA and its variants have established strong baselines by connecting CLIP vision encoders to language models through learnable projectors [?]. More recent works like Qwen2-VL [?], Cambrian-1 [?], and POINTS [?] have explored various architectural improvements, training strategies, and data scaling approaches. However, most existing MLLMs still struggle with complex reasoning tasks that require systematic step-by-step analysis.

Visual Reasoning Benchmarks. The evaluation of visual reasoning capabilities has been advanced by several comprehensive benchmarks. MMMU [?] provides a massive multi-discipline evaluation covering diverse expert-level reasoning tasks, while MMMU-Pro [?] offers a more robust assessment with reduced data contamination. MathVista [?] focuses specifically on mathematical reasoning in visual contexts, and ChartQA [?] evaluates reasoning about charts and graphs. These benchmarks have revealed significant gaps in current MLLMs' reasoning capabilities, motivating the development of specialized approaches.

Chain-of-Thought Reasoning. Chain-of-thought (CoT) prompting has shown remarkable effectiveness in enhancing language models' reasoning capabilities [?]. However, direct application of CoT to multimodal settings faces unique challenges. Visual reasoning often requires more complex integration of perceptual and logical processing, and the generation of high-quality reasoning chains for visual content is significantly more challenging than for text-only scenarios. Previous attempts to apply CoT to MLLMs have shown mixed results, often leading to instability and degraded perception capabilities.

Direct Preference Optimization. Direct Preference Optimization (DPO) [?] has emerged as a powerful alternative to reinforcement learning from human feedback (RLHF) for aligning language models with human preferences. DPO directly optimizes the policy model using preference data without requiring a separate reward model, making it more stable and efficient than RLHF. Recent works have begun exploring DPO for multimodal settings, but its application to long-chain visual reasoning remains underexplored.

Multi-Agent Systems. The decomposition of complex tasks into specialized agents has shown promise in various AI applications. In the context of reasoning, separating different aspects of the problem-solving process can improve both performance and interpretability. Our multi-agent approach draws inspiration from this line of work, specifically targeting the separation of reasoning generation and answer synthesis in visual reasoning tasks.

3. Methodology

Our Insight-V framework addresses the challenges of long-chain visual reasoning through three key components: (1) a scalable data generation pipeline that produces high-quality structured reasoning chains, (2) a multi-agent training framework that separates reasoning from answering, and (3) an iterative preference optimization procedure that refines reasoning quality. Figure 1 provides an overview of our approach.

3.1. Scalable Data Generation Pipeline

The foundation of effective long-chain visual reasoning lies in high-quality training data that demonstrates systematic reasoning processes. We develop a progressive data generation pipeline that creates structured reasoning chains with explicit control mechanisms.

Progressive Reasoning Generator. Our reasoning generator employs a structured JSON format with explicit action indicators to control the reasoning flow. Each reasoning step

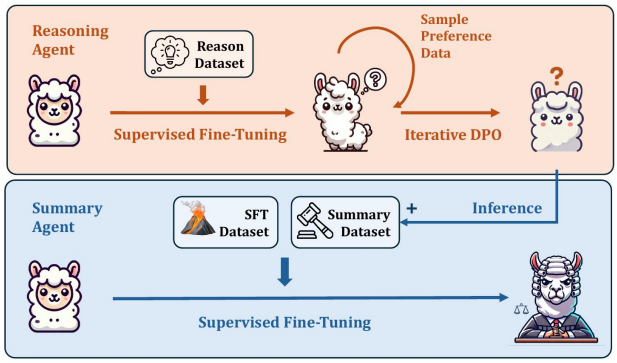

Figure 1. **Overview of Insight-V Model Design.** Our framework employs a multi-agent approach with separate reasoning and summary agents, trained through supervised fine-tuning and iterative DPO.

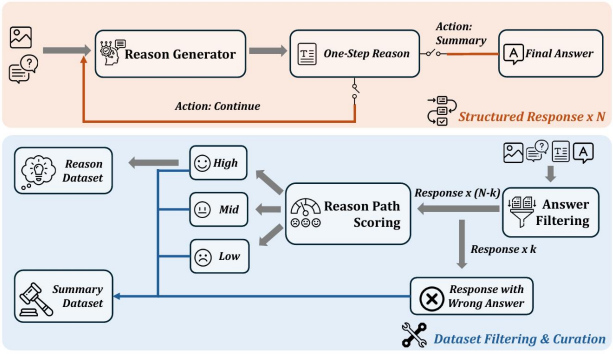

Figure 2. **Data Generation Pipeline of Insight-V.** Our pipeline uses progressive generation to create diverse reasoning paths, followed by multi-granularity assessment to filter and rank the generated chains.

includes an action field that can be either "continue" (to generate another reasoning step) or "summary" (to conclude the reasoning chain). This design enables the generation of diverse reasoning paths of varying lengths for the same query, providing rich training supervision for different reasoning strategies.

The generator is prompted to think step-by-step about visual content, breaking down complex problems into manageable sub-problems. For mathematical reasoning tasks, it might identify relevant visual elements, extract numerical information, establish relationships between quantities, and perform calculations. For chart analysis tasks, it could involve trend identification, data extraction, comparative analysis, and conclusion synthesis.

Multi-Granularity Assessment. To ensure data quality, we implement a two-stage assessment procedure. First, we use a strong language model (e.g., Qwen2) to filter out reasoning chains that lead to incorrect answers, eliminating obviously flawed reasoning early in the process. Second, we employ a capable MLLM (e.g., Qwen2-VL) to score the remaining reasoning chains based on their logical consistency, clarity, and relevance to the visual content.

This multi-granularity approach allows us to maintain high standards for both correctness and reasoning quality while scaling to large datasets. The scoring mechanism considers factors such as step-by-step logical flow, appropriate use of visual information, mathematical accuracy, and overall coherence of the reasoning chain.

3.2. Multi-Agent Training Framework

Traditional approaches to training reasoning capabilities often entangle the reasoning process with answer generation, leading to instability and error propagation. We address this by decomposing the task into two specialized agents with distinct responsibilities.

Reasoning Agent. The reasoning agent is trained exclusively on generating high-quality reasoning chains. It receives visual input and questions, then produces structured reasoning in JSON format with explicit step-by-step analysis. The agent is trained using supervised fine-tuning on approximately 200K of the highest-quality reasoning chains identified through our data generation pipeline.

The reasoning agent learns to systematically approach visual reasoning problems by identifying key visual elements, establishing relevant relationships, and building logical arguments step by step. Importantly, this agent does not generate final answers, allowing it to focus solely on the reasoning process without being influenced by answer-specific biases.

Summary Agent. The summary agent specializes in synthesizing final answers based on reasoning content. It receives the same visual input and question as the reasoning agent, but also gets access to reasoning chains through special `<thoughts>` tags. This design allows the summary agent to either utilize the provided reasoning or ignore it when the reasoning quality is poor.

The summary agent is trained on approximately 1.2M examples that include a mixture of optimal reasoning paths, suboptimal paths, and general question-answering data without reasoning chains. This diverse training regime ensures that the agent maintains strong perception capabilities while learning to effectively leverage reasoning when available.

3.3. Iterative Preference Optimization

To further refine the quality of reasoning chains and improve the stability of the multi-agent system, we employ an iterative Direct Preference Optimization (DPO) procedure.

Preference Data Generation. For each query in our dataset, we generate multiple reasoning chains using the current reasoning agent with different sampling parameters. We then rank these chains using our multi-granularity assessment

pipeline, creating preference pairs that capture relative quality differences between reasoning chains.

Iterative DPO Training. We perform three rounds of DPO training, with each round using approximately 15K preference pairs. The learning rate is set to 5e-7 to ensure stable convergence. After each DPO round, we regenerate preference pairs using the updated model, allowing for continuous improvement in reasoning quality.

This iterative approach enables the reasoning agent to progressively learn more sophisticated reasoning patterns while maintaining stability. The preference-based training naturally encourages longer, more detailed reasoning chains when they lead to better outcomes, while discouraging verbose but unhelpful reasoning.

3.4. Model Implementation Details

We apply our Insight-V framework to two base models: LLaVA-NeXT-8B [?] with CLIP ViT-L/14 vision encoder at 336px resolution and a 2-layer MLP connector, and a custom Qwen2.5-7B based MLLM pretrained on 558K captions and 4M supervised fine-tuning examples.

The training procedure follows a carefully designed curriculum: first, we train the reasoning agent using supervised fine-tuning on high-quality reasoning chains, then we apply iterative DPO for reasoning refinement. Simultaneously, we train the summary agent on the diverse mixture of reasoning and non-reasoning examples. The final system integrates both agents during inference, with the reasoning agent providing structured analysis that the summary agent can leverage for final answer generation.

4. Experiments

We conduct comprehensive experiments to evaluate the effectiveness of Insight-V across multiple dimensions: visual reasoning capabilities, general perception performance, and ablation studies on key design choices.

4.1. Experimental Setup

Benchmarks. We evaluate visual reasoning capabilities on seven challenging benchmarks: MMMU [?], MMMU-Pro [?], MMBench, MME, ChartQA [?], MMStar, and MathVista [?]. For general perception, we use TextVQA, DocVQA, OCRBench, and AI2D. All evaluations use the lmms-eval framework, with MathVista specifically judged by GPT-4-0613 for consistency with prior work.

Baselines. We compare against state-of-the-art MLLMs including DeepSeek-VL, VILA-1.5, Cambrian-1 [?], InternLM-XComposer2, POINTS [?], IXC-2.5, Bunny-LLaMA3, MM-1.5, MiniCPM series, Qwen2-VL [?], Idefics3-LLaMA3, and Ovis1.5-LLaMA3.

Implementation Details. Our reasoning agent is trained with supervised fine-tuning on 200K high-quality reasoning chains, followed by 3 rounds of iterative DPO with 15K preference pairs per round (learning rate 5e-7). The summary agent is trained on 1.2M examples mixing optimal/flawed reasoning paths and general QA data. We use the same training hyperparameters as the base models for consistency.

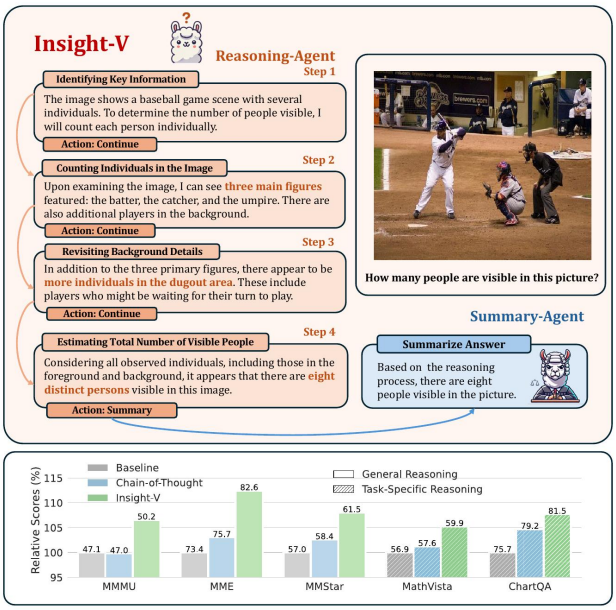

Figure 3. **Illustration and Performance of Insight-V.** Our multi-agent approach shows clear improvements in visual reasoning tasks while maintaining strong perception capabilities.

4.2. Main Results

Table 1 presents our main results on visual reasoning benchmarks. Insight-V demonstrates substantial and consistent improvements across all evaluated models and benchmarks.

LLaVA-NeXT Results. Applied to LLaVA-NeXT-8B, Insight-V achieves an average improvement of +7.0 points (40.2→47.2). The improvements are particularly notable on challenging reasoning benchmarks: MMMU improves from 36.9 to 42.0, MMMU-Pro shows a dramatic increase from 13.2 to 21.0, and MMBench gains from 72.3 to 81.7. ChartQA improves from 69.4 to 77.4, MMStar from 43.1 to 57.4, and MathVista from 45.9 to 49.8. Even on the perception-heavy MME benchmark, we observe improvements on both sub-tasks.

Stronger Base Model Results. On our custom 7B baseline model, Insight-V achieves consistent improvements of +2.9 points average (48.7→51.6). This demonstrates the generalizability of our approach across different model architectures and scales. Notable improvements include MMMU (47.1→50.2), MMMU-Pro (22.6→24.9), MMBench (81.3→82.3), ChartQA (75.7→81.5), MMStar (57.0→61.5), and MathVista (56.9→59.9).

Table 1. **Results on Visual Reasoning Tasks.** We conduct evaluation experiments across 7 benchmarks, covering both general reasoning and task-specific reasoning assessments. Insight-V exhibits notable effectiveness and generalizability when applied to LLaVA-NeXT and our baseline model, surpassing other state-of-the-art MLLMs by a large margin.

Model	Size	MMMU	MMMU-Pro	MMBench	MME	ChartQA	MMStar	MathVista	Average
DeepSeek-VL [?]	7B	35.4	-	73.5	-/-	59.1	37.1	36.1	-
VILA-1.5 [?]	8B	38.6	-	75.3	1634.9/-	-	39.7	-	-
Cambrian-1 [?]	8B	42.7	-	75.9	1547.1/-	73.3	-	49.0	-
InternLM-XComposer2 [?]	7B	41.1	-	77.6	2220.4	71.8	56.2	59.5	-
POINTS [?]	7B	51.4	-	78.0	2184.1	-	60.9	63.0	-
IXC-2.5 [?]	7B	42.9	-	79.4	2233.1	82.2	59.9	**63.7**	-
Bunny-LLaMA3 [?]	8B	43.4	-	77.2	1588.9/321.1	-	-	34.4	-
MM-1.5 [?]	7B	41.8	-	-	1514.9/346.4	78.6	-	47.6	-
MiniCPM-LLaMA3-V 2.5 [?]	8B	45.8	19.6	77.2	2024.6	-	51.8	54.3	-
MiniCPM-V-2.6 [?]	7B	49.8	**27.2**	78.0	2268.7	-	57.5	60.6	-
Qwen2-VL [?]	7B	**53.7**	-	81.0	-	**83.0**	60.7	61.4	-
Idefics3-LLaMA3 [?]	8B	46.6	22.9	77.5	1937.4	74.8	55.9	58.4	48.1
Ovis1.5-LLaMA3 [?]	8B	48.3	23.6	76.6	1948.5	76.4	57.3	63.0	49.4
LLaVA-NeXT-LLaMA3 [?]	8B	36.9	13.2	72.3	1611.1/346.0	69.4	43.1	45.9	40.2
+ Multi-Agent	8B	40.8	17.8	77.6	1603.7/469.3	74.6	52.6	47.4	44.5
+ Iterative DPO (**Insight-V-LLaVA**)	8B	42.0	21.0	81.7	1583.9/485.4	77.4	57.4	49.8	**47.2** (+7.0)
Our Base Model	7B	47.1	22.6	81.3	1573.7/482.5	75.7	57.0	56.9	48.7
+ Multi-Agent	7B	49.7	23.8	82.2	1662.2/**629.3**	81.2	58.6	58.7	50.7
+ Iterative DPO (**Insight-V**)	7B	50.2	24.9	**82.3**	**1685.1**/627.0	81.5	**61.5**	59.9	**51.6** (+2.9)

Table 2. **Results on other multimodal benchmarks.** Insight-V enhances reasoning capabilities without compromising general visual perception and even achieves improvements on benchmarks requiring perception ability more.

Model	TextVQA	DocVQA	OCRBench	AI2D
LLaVA-NeXT-LLaMA3	65.2	78.2	553	71.5
+ Multi-Agent	68.9	81.8	631	75.7
+ Iterative DPO (**Insight-V-LLaVA**)	**70.5**	**82.9**	**663**	**77.3**
Our Base Model	75.4	90.2	713	79.7
+ Multi-Agent	**77.0**	91.4	**738**	**80.1**
+ Iterative DPO (**Insight-V**)	76.8	**91.5**	735	79.8

Comparison with State-of-the-Art. Our results significantly outperform existing approaches. Insight-V-LLaVA achieves competitive performance with much larger models and specialized approaches. Notably, our 8B model achieves better average performance than many 7B+ specialized models like POINTS, IXC-2.5, and Ovis1.5-LLaMA3.

4.3. General Perception Results

Table 2 demonstrates that Insight-V enhances reasoning capabilities without compromising general visual perception. In fact, we observe improvements across multiple perception-focused benchmarks.

For LLaVA-NeXT-8B, we see significant improvements on TextVQA (65.2→70.5), DocVQA (78.2→82.9), OCRBench (553→663), and AI2D (71.5→77.3). This suggests that the structured reasoning training actually helps with tasks that require careful visual analysis and text understanding.

Our stronger 7B baseline maintains its strong perception capabilities while gaining reasoning abilities. The summary agent's training on diverse data ensures that general visual understanding is preserved or enhanced.

Table 3. **Ablations on the Insight-V Design Choice.** The multi-agent design outperforms other configurations, highlighting the critical role of reasoning and summarization decomposition.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Baseline	47.1	75.7	56.9	57.0	59.2
Vanilla - Direct SFT	47.0	79.2	57.6	58.4	60.6
Multi-Turn Supervised	48.1	79.6	57.9	58.2	61.0
Summary Agent Only	47.5	76.3	57.3	57.9	59.8
Multi-Agent	49.7	81.2	58.7	58.6	62.1

4.4. Ablation Studies

We conduct comprehensive ablation studies to validate our design choices and understand the contribution of each component.

Multi-Agent Design Analysis. Table 3 shows the effectiveness of our multi-agent approach compared to alternative designs. The multi-agent framework (62.1 average) significantly outperforms direct supervised fine-tuning on reasoning chains (60.6), multi-turn supervised learning (61.0), and using only the summary agent (59.8). This validates our core hypothesis that separating reasoning from answering improves overall performance.

DPO Strategy Analysis. Table 4 demonstrates the benefits of our iterative DPO approach. Iterative DPO (63.3 average) outperforms single-pass DPO (62.7), RLAIF (62.3), and the multi-agent baseline without preference optimization (62.1). The iterative nature allows for progressive refinement of reasoning quality.

Data Scaling Analysis. Figure 4 shows the relationship between training data amount and performance across different benchmarks. Performance consistently improves with more high-quality reasoning data, with most benchmarks showing

Table 4. **Ablations on the DPO training strategy.** Iterative DPO progressively enhances the model's reasoning capabilities, leading to improved performance.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Insight-V (Multi-Agent)	49.7	81.2	58.7	58.6	62.1
+ RLAIF	49.5	81.4	59.1	59.2	62.3
+ DPO	50.8	80.8	59.3	59.9	62.7
+ Iterative DPO	50.2	81.5	59.9	61.5	63.3

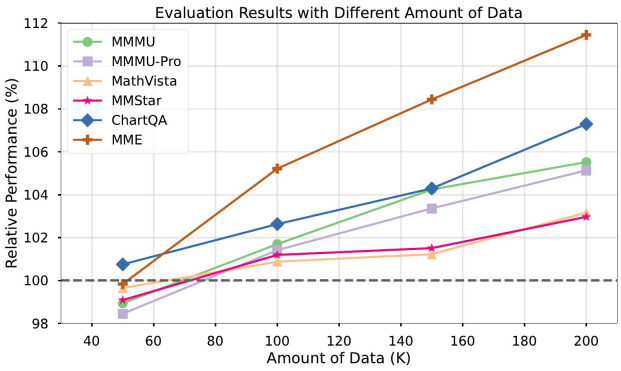

Figure 4. **Ablations on the amount of training data.** Performance consistently improves with more high-quality reasoning data across multiple benchmarks.

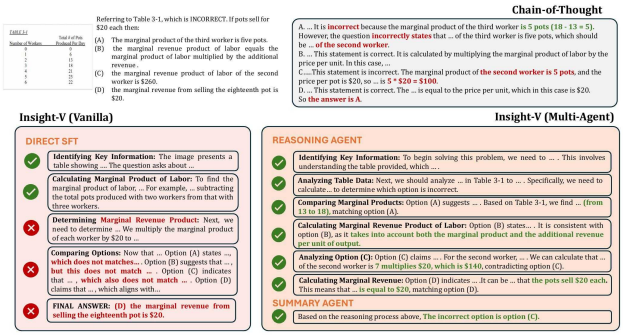

Figure 5. **Qualitative Results of Insight-V.** Our multi-agent approach produces more structured and comprehensive reasoning chains compared to vanilla models.

continued gains up to our full dataset of 200K examples.

4.5. Qualitative Analysis

Figure 5 presents qualitative examples comparing Insight-V outputs with baseline models. Our approach generates more structured, systematic reasoning that better utilizes visual information and follows logical progression.

The reasoning agent produces detailed step-by-step analysis that identifies relevant visual elements, establishes relationships, and builds toward conclusions. The summary agent then synthesizes this information into clear, accurate answers. This decomposition leads to more reliable and interpretable reasoning processes.

4.6. Error Analysis and Limitations

While Insight-V shows substantial improvements, some limitations remain. Complex multi-step mathematical reasoning still occasionally suffers from error propagation, though less frequently than baseline approaches. The multi-agent framework adds computational overhead during inference, requiring careful optimization for deployment scenarios.

The quality of generated reasoning chains depends heavily on the base model's capabilities, and the iterative DPO procedure requires careful hyperparameter tuning to avoid overfitting to specific reasoning patterns.

5. Conclusion

We presented Insight-V, a comprehensive framework that significantly advances long-chain visual reasoning capabilities in Multimodal Large Language Models. Our key insight is that separating reasoning from answering through specialized agents, combined with high-quality structured training data and iterative preference optimization, leads to substantial improvements in visual reasoning while maintaining strong general perception capabilities.

Our contributions include: (1) a scalable automated pipeline for generating high-quality long-chain visual reasoning data through progressive generation and multi-granularity assessment, (2) a multi-agent MLLM framework that disentangles reasoning and summarization processes, improving robustness to error propagation, (3) an iterative DPO procedure that progressively enhances reasoning quality through model-generated preferences, and (4) comprehensive experimental validation demonstrating consistent improvements across diverse visual reasoning benchmarks.

Experimental results show that Insight-V achieves substantial improvements on challenging visual reasoning tasks: +7.0 points average on LLaVA-NeXT-8B (40.2→47.2) and +2.9 points on our 7B baseline (48.7→51.6). Notably, these gains come without sacrificing general visual perception capabilities, with improvements observed on TextVQA, DocVQA, OCRBench, and AI2D.

The success of Insight-V demonstrates the importance of structured training data and careful task decomposition in developing capable visual reasoning systems. The multi-agent approach provides a principled way to scale reasoning capabilities while maintaining stability and interpretability.

Future Directions. Our work opens several promising research directions. The progressive data generation pipeline could be extended to other modalities and reasoning types. The multi-agent framework could benefit from more sophisticated reasoning-summary interaction mechanisms. Additionally, exploring the integration of external knowledge sources and tool use within the reasoning chains could further enhance capabilities.

The scalability and generalizability of Insight-V across

different model architectures suggest that our approach could serve as a foundation for developing more capable multi-modal reasoning systems. As visual reasoning becomes increasingly important for real-world applications, frameworks like Insight-V provide essential stepping stones toward more sophisticated and reliable AI systems that can truly understand and reason about visual content.